

Министерство науки и высшего образования Российской Федерации
НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ
ТОМСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ (НИ ТГУ)
Институт прикладной математики и информатики

ОТЧЕТ ПО ЛАБОРАТОРНОЙ РАБОТЕ
по дисциплине «Системы управления базами данных»

Подзапросы

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«26» ноября 2022 г.

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«26» ноября 2022 г.

1. Вывести фамилию сотрудника, его зарплату и среднюю зарплату в компании.
 SELECT last_name, salary, (SELECT AVG(salary) FROM HR.employees) AS "AVG SALARY" FROM HR.employees;

Worksheet

Query Builder

```
SELECT last_name, salary, (SELECT AVG(salary) FROM HR.employees) AS "AVG SALARY"
FROM HR.employees;
```

 Query Result x

    SQL | Fetched 50 rows in 0,004 seconds

	LAST_NAME	SALARY	AVG SALARY
1	OConnell	2600	6461,831775700934579439252336448598130841
2	Grant	2600	6461,831775700934579439252336448598130841
3	Whalen	4400	6461,831775700934579439252336448598130841
4	Hartstein	13000	6461,831775700934579439252336448598130841
5	Fay	6000	6461,831775700934579439252336448598130841
6	Mavris	6500	6461,831775700934579439252336448598130841

2. Вывести номер отдела и зарплату, которая находится в промежутке от 2000 до 3000, через подзапрос в FROM.
 SELECT * FROM (SELECT salary, department_id FROM HR.employees WHERE salary BETWEEN 2000 and 3000);

Worksheet	Query Builder
<pre>SELECT * FROM (SELECT salary, department_id FROM HR.employees WHERE salary BETWEEN 2000 and 3000);</pre>	
Query Result x All Rows Fetched: 26 in 0,003 seconds	
SALARY	DEPARTMENT_ID
1 2600	50
2 2600	50
3 2900	30
4 2800	30
5 2600	30
6 2500	30

3. Вернуть имена сотрудников с такой же фамилией, как у Smith.
 SELECT last_name, first_name FROM HR.employees WHERE first_name = ANY (SELECT first_name FROM HR.employees WHERE INITCAP (last_name) = 'Smith');

Worksheet	Query Builder
<pre>SELECT last_name, first_name FROM HR.employees WHERE first_name = ANY (SELECT first_name FROM HR.employees WHERE INITCAP (last_name) = 'Smith');</pre>	
Query Result x All Rows Fetched: 3 in 0,004 seconds	
LAST_NAME	FIRST_NAME
1 Smith	Lindsey
2 Gietz	William
3 Smith	William

4. Вернуть имена и фамилии сотрудников отдела, где работает Tobias.

Worksheet

Query Builder

```
SELECT last_name, first_name FROM HR.employees WHERE department_id =
(SELECT department_id FROM HR.employees WHERE INITCAP(last_name) = 'Tobias');
```

Query Result x

All Rows Fetched: 6 in 0,003 seconds

	LAST_NAME	FIRST_NAME
1	Raphaely	Den
2	Khoo	Alexander
3	Baida	Shelli
4	Tobias	Sigal
5	Himuro	Guy
6	Colmenares	Karen

- ```
SELECT last_name, first_name, salary FROM HR.employees WHERE salary < (SELECT AVG (salary) FROM HR.employees);
```

Worksheet

Query Builder

```
SELECT last_name, first_name, salary FROM HR.employees
WHERE salary < (SELECT AVG (salary) FROM HR.employees);
```

Query Result x

SQL | Fetched 50 rows in 0,003 seconds

|   | LAST_NAME | FIRST_NAME | SALARY |
|---|-----------|------------|--------|
| 1 | OConnell  | Donald     | 2600   |
| 2 | Grant     | Douglas    | 2600   |
| 3 | Whalen    | Jennifer   | 4400   |
| 4 | Fay       | Pat        | 6000   |
| 5 | Ernst     | Bruce      | 6000   |
| 6 | Austin    | David      | 4800   |

- ```
SELECT department_id, AVG(salary) FROM HR.employees GROUP BY department_id HAVING
AVG(salary) > (SELECT AVG(salary) FROM HR.employees WHERE department_id = 30);
```

[illegible]

- 3

[illegible]

- ```
SELECT last_name, first_name, department_id FROM HR.employees WHERE department_id IN (SELECT
department_id FROM HR.departments WHERE department_name = 'IT' or location_id = 1800);
```

Worksheet

Query Builder

```

SELECT last_name, first_name, department_id FROM HR.employees
WHERE department_id IN (SELECT department_id FROM HR.departments
WHERE department_name = 'IT' or location_id = 1800);

```

Query Result x

|   | LAST_NAME | FIRST_NAME | DEPARTMENT_ID |
|---|-----------|------------|---------------|
| 1 | Hartstein | Michael    | 20            |
| 2 | Fay       | Pat        | 20            |
| 3 | Hunold    | Alexander  | 60            |
| 4 | Ernst     | Bruce      | 60            |
| 5 | Austin    | David      | 60            |
| 6 | Pataballa | Valli      | 60            |
| 7 | Lorentz   | Diana      | 60            |

- ```
SELECT E.first_name, D.department_name FROM HR.employees E LEFT JOIN HR.departments D
USING(department_id) WHERE E.salary = (SELECT salary FROM HR.employees WHERE
UPPER(first_name) = 'SAMUEL');
```

Worksheet

Query Builder

```

SELECT E.first_name, D.department_name FROM HR.employees E LEFT
JOIN HR.departments D USING(department_id) WHERE E.salary =
(SELECT salary FROM HR.employees WHERE UPPER(first_name) = 'SAMUEL');

```

Query Result x

| All Rows Fetched: 4 in 0,004 seconds

	FIRST_NAME	DEPARTMENT_NAME
1	Julia	Shipping
2	Stephen	Shipping
3	Winston	Shipping
4	Samuel	Shipping

- 4

[illegible]

```
SELECT E.first_name, J.job_title FROM HR.employees E JOIN HR.jobs J USING(job_id) WHERE
E.department_id = (SELECT department_id FROM HR.employees WHERE first_name = 'Alana');
```

Worksheet

Query Builder

```

SELECT E.first_name, J.job_title FROM HR.employees E JOIN HR.jobs J
USING(job_id) WHERE E.department_id = (SELECT department_id FROM HR.employees
WHERE first_name = 'Alana');

```

Query Result x

SQL

All Rows Fetched: 45 in 0,005 seconds

	FIRST_NAME	JOB_TITLE
1	Donald	Shipping Clerk
2	Kevin	Shipping Clerk
3	Douglas	Shipping Clerk
4	Alana	Shipping Clerk
5	Vance	Shipping Clerk
6	Samuel	Shipping Clerk

```
SELECT department_id, first_name, last_name FROM HR.employees WHERE employee_id = SOME
(SELECT manager_id FROM HR.departments);
```

Worksheet

Query Builder

```

SELECT department_id, first_name, last_name FROM HR.employees
WHERE employee_id = SOME (SELECT manager_id FROM HR.departments);

```

Query Result x

	DEPARTMENT_ID	FIRST_NAME	LAST_NAME
1	10	Jennifer	Whalen
2	20	Michael	Hartstein
3	30	Den	Raphaely
4	40	Susan	Mavris
5	50	Adam	Fripp
6	60	Alexander	Hunold
7	70	Hermann	Baer
8	80	John	Russell
9	90	Steven	King
10	100	Nancy	Greenberg
11	110	Shelley	Higgins

```
SELECT department_id, first_name, last_name FROM HR.employees WHERE employee_id = ANY (SELECT
manager_id FROM HR.departments);
```

Worksheet

Query Builder

```
SELECT department_id, first_name, last_name FROM HR.employees
WHERE employee_id = ANY (SELECT manager_id FROM HR.departments);
```

Query Result x

SQL | All Rows Fetched: 11 in 0,004 seconds

	DEPARTMENT_ID	FIRST_NAME	LAST_NAME
1	10	Jennifer	Whalen
2	20	Michael	Hartstein
3	30	Den	Raphaely
4	40	Susan	Mavris
5	50	Adam	Fripp
6	60	Alexander	Hunold
7	70	Hermann	Baer
8	80	John	Russell
9	90	Steven	King
10	100	Nancy	Greenberg
11	110	Shelley	Higgins

14. Вывести имя, фамилию и зарплату сотрудника, зарплата которого больше всех средних зарплат по отделениям.

SELECT first_name, last_name, salary FROM HR.employees WHERE salary > ALL (SELECT AVG(salary) FROM HR.employees GROUP BY department_id);

Worksheet

Query Builder

```
SELECT first_name, last_name, salary FROM HR.employees WHERE salary > ALL (SELECT AVG(salary) FROM HR.employees GROUP BY department_id);
```

▶

Query Result x



SQL | All Rows Fetched: 1 in 0,004 seconds

	FIRST_NAME	LAST_NAME	SALARY
1	Steven	King	24000

15. Вывести фамилию, имя, номер отделения и зарплату сотрудника, зарплата которого больше чем средняя в его отделении.

SELECT department_id, first_name, last_name, salary FROM HR.employees E1 WHERE salary > (SELECT AVG(salary) FROM HR.employees WHERE E1.department_id = department_id);

Worksheet

Query Builder

```
SELECT department_id, first_name, last_name, salary FROM HR.employees E1
WHERE salary > (SELECT AVG(salary) FROM HR.employees
WHERE E1.department_id = department_id);
```

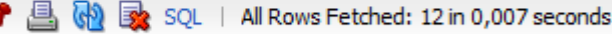
Query Result x

| All Rows Fetched: 38 in 0,008 seconds

	DEPARTMENT_ID	FIRST_NAME	LAST_NAME	SALARY
1	20	Michael	Hartstein	13000
2	110	Shelley	Higgins	12008
3	90	Steven	King	24000
4	60	Alexander	Hunold	9000
5	60	Bruce	Ernst	6000
6	100	Nancy	Greenberg	12008

16. Вывести фамилии, имена, номера отделений и зарплату сотрудников, зарплата которых является минимальной в их отделении.

SELECT department_id, first_name, last_name, salary FROM HR.employees WHERE (salary, department_id) IN (SELECT MIN(salary), department_id FROM HR.employees GROUP BY department_id);

Worksheet		Query Builder		
		<pre>SELECT department_id, first_name, last_name, salary FROM HR.employees WHERE (salary, department_id) IN (SELECT MIN(salary), department_id FROM HR.employees GROUP BY department_id);</pre>		
		Query Result x  		
	DEPARTMENT_ID	FIRST_NAME	LAST_NAME	SALARY
1	50	TJ	Olson	2100
2	10	Jennifer	Whalen	4400
3	20	Pat	Fay	6000
4	40	Susan	Mavris	6500
5	70	Hermann	Baer	10000
6	110	William	Gietz	8300
7	90	Neena	Kochhar	17000
8	90	Lex	De Haan	17000
9	60	Diana	Lorentz	4200
10	100	Luis	Popp	6900
11	30	Karen	Colmenares	2500
12	80	Sundita	Kumar	6100